A CLOSED SUBSCHEME OF PROJECTIVE SPACE OVER A NOETHERIAN UFD ALL OF WHOSE COMPONENTS HAVE CODIMENSION ONE IS CUT OUT BY A SINGLE HOMOGENEOUS POLYNOMIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 8041-8049 and proof span 8051-8091 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof displays the dehomogenized equation on each standard open and the divisibility relations after clearing denominators; both displays are preserved unchanged.

Editorial note. The author writes the ideal of the subscheme as the kernel of the map to the sum of the twists of the structure sheaf's sections; that description is kept as written.

Editorial note. 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The polynomial ring over a unique factorisation domain is a unique factorisation domain, cited from the algebra chapter. Cited at proof lines 8053-8056 and again at 8084-8085. Shows that the local rings of projective space over the base are unique factorisation domains, and later allows the divisibility argument on homogeneous polynomials.
- A closed subscheme without embedded points all of whose components have codimension one in a scheme with unique factorisation domain local rings is an effective Cartier divisor, cited from the same chapter. Cited at proof lines 8057-8058. Turns the given closed subscheme into an effective Cartier divisor, hence its ideal sheaf is invertible.
- On projective space over a unique factorisation domain every invertible sheaf is a twist of the structure sheaf, cited from the same chapter. Cited at proof lines 8060-8062. Identifies the invertible ideal sheaf with a twist of the structure sheaf.

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- The description of sections of twists of the structure sheaf on projective space over a ring as homogeneous polynomials of the corresponding degree, cited from the cohomology of schemes chapter. Cited at proof lines 8066-8068. Produces the single homogeneous polynomial cutting out the divisor on each standard open.

Source reference. Original lemma 0BC1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14264–14264. Same-chapter reference used in the retained author text at source line(s) 7201, 8055. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01XT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2532–2532. Same-chapter reference used in the retained author text at source line(s) 8071. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0BXJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14982–14982. Same-chapter reference used in the retained author text at source line(s) 8062. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0BXH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14980–14980. Same-chapter reference used in the retained author text at source line(s) 3187, 8057. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let R be a Noetherian UFD. Let $Z \subset \mathbf{P}^n_R$ be a closed subscheme which has no embedded points such that every irreducible component of Z has codimension 1 in $\mathbf{P}^n_R$. Then the ideal $I(Z) \subset R[T_0, \ldots, T_n]$ corresponding to Z is principal.*

Proof. Observe that the local rings of $X = \mathbf{P}^n_R$ are UFDs because X is covered by affine pieces isomorphic to $\mathbf{A}^n_R$ and $R[x_1, \ldots, x_n]$ is a UFD (Algebra, Lemma 0BC1). Thus Z is an effective Cartier divisor by Lemma 0BXH. Let $\mathcal{I} \subset \mathcal{O}_X$ be the quasi-coherent sheaf of ideals corresponding to Z. Choose an isomorphism $\mathcal{O}(m) \to \mathcal{I}$ for some $m \in \mathbf{Z}$, see Lemma 0BXJ. Then the composition

$$\mathcal{O}_X(m) \to \mathcal{I} \to \mathcal{O}_X$$

is nonzero. We conclude that $m \leq 0$ and that the corresponding section of $\mathcal{O}_X(m)^{\otimes -1} = \mathcal{O}_X(-m)$ is given by some $F \in R[T_0, \ldots, T_n]$ of degree $-m$, see Cohomology of Schemes, Lemma 01XT. Thus on the ith standard open $U_i = D_+(T_i)$ the closed subscheme $Z \cap U_i$ is cut out by the ideal

$$(F(T_0/T_i, \ldots, T_n/T_i)) \subset R[T_0/T_i, \ldots, T_n/T_i]$$

Thus the homogeneous elements of the graded ideal $I(Z) = \mathrm{Ker}(R[T_0, \ldots, T_n] \to \bigoplus \Gamma(\mathcal{O}_Z(m)))$ is the set of homogeneous polynomials G such that

$$G(T_0/T_i, \ldots, T_n/T_i) \in (F(T_0/T_i, \ldots, T_n/T_i))$$

for $i = 0, \ldots, n$. Clearing denominators, we see there exist $e_i \geq 0$ such that

$$T_i^{e_i} G \in (F)$$

for $i = 0, \ldots, n$. As R is a UFD, so is $R[T_0, \ldots, T_n]$. Then $F | T_0^{e_0} G$ and $F | T_1^{e_1} G$ implies $F | G$ as $T_0^{e_0}$ and $T_1^{e_1}$ have no factor in common. Thus $I(Z) = (F)$. $\square$